

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)		1. Carbon in the (molecules of) dead insects/ant faeces (1) 2. Ref to role of decomposers/decomposed by {bacteria/fungi} (1) 3. Carbon dioxide released (1) 4. from respiration (1) Accept respiration of ants or decomposers for MP3 and MP4	2	2		4		
		(ii)		Any 3 × (1) from: {Decomposers/bacteria/fungi} release ammonium ions/ NH_4^+ ammonification/putrefaction (1) Ammonium/ NH_4^+ converted to nitrite/ NO_2^- (1) Reject ammonia Nitrite/ NO_2^- converted to nitrate/ NO_3^- (1) By (both named) nitrifying bacteria/nitrification (1) If allocate named bacteria to each reaction, must be correct	3			3		
	(b)	(i)		Any 2 × (1) from: less CO_2 used in photosynthesis (1) More CO_2 produced in combustion (1) More decomposition so more CO_2 (1)		2		2		
		(ii)		Less respiration (by plants/animals/decomposers)		1		1		
				Question 3 total	5	5	0	10	0	0